Special Summer Integrated Learning Programme

Mathematics

Introduction to Integral Calculus

Week 6 Day 1 PM

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Outline

Intuitions

Integration is the reverse process of differentiation.

$$\begin{tikzcd} f(x) \arrow[rr, bend left, "\text{differentiation}"] & & f'(x) \arrow[ll, bend left, "\text{integration}"] \end{tikzcd}$$

E.g.:

$$\begin{tikzcd} \text{Displacement } s(t) \arrow[rr, bend left, "\text{differentiation}"] & & \text{Velocity } v(t) \arrow[ll, bend left, "\text{integration}"] \arrow[rr, bend left, "\text{differentiation}"] & & \text{Acceleration } a(t) \arrow[ll, bend left, "\text{integration}"] \end{tikzcd}$$

Antiderivatives

Recall that differentiation defines a new function by finding the instantaneous rate of change of a given function.

Function $F(x)$ $\underset{\text{Differentiate}}{\longrightarrow}$ Derivative $F'(x)$

Example:

$F_1(x) = 3x^4 + x$ $\underset{\text{Differentiate}}{\longrightarrow}$ $F_1'(x) = 12x^3 + 1$

$F_2(x) = 3x^4 + x + 10$ $\underset{\text{Differentiate}}{\longrightarrow}$ $F_2'(x) = 12x^3 + 1$

Note that in general, two different functions can have the same derivative.

Definition:

If $f(x)$ is given, then a function $F(x)$ is called an ***antiderivative*** of $f(x)$ if $F'(x) = f(x)$.

Note that f is an antiderivative of f', f' is the derivative of f.

Indefinite integrals

What is the set of all functions with a given derivative? I.e. the set of all antiderivatives of a given function?

Theorem

If $f'(x) = g'(x)$, then $f(x) = g(x) + c$ for some constant c.

Proof: $(f(x) - g(x))' = f'(x) - g'(x) = 0$. Hence $f(x) - g(x)$ is a constant c and therefore $f(x) = g(x) + c$. Q.E.D.

Theorem

Suppose $F(x)$ is an antiderivative of $f(x)$, then the set of all antiderivatives of $f(x)$ is

$$\{F(x) + c \mid c \in \mathbb{R} \text{ is any constant}\}$$

Indefinite integrals

The previous theorems are important since they show that if we try to find a function based on its derivative, we can only find the original function up to a constant. This is integration!!

Definition:

The ***indefinite integral*** of $f(x)$, denoted as $\int f(x)\,\mathrm{d}x$ is the general antiderivative of $f(x)$, i.e.

$$\int f(x)\,\mathrm{d}x = F(x) + c,$$

where $F(x)$ is an antiderivative of $f(x)$ and c is an arbitrary constant. The process of computing $F(x) + c$ from $f(x)$ is called ***integration***. The verb is ***to integrate***.

Since we don't get a definite answer from integration, that's why it's called an indefinite integral.

Indefinite integrals - example

E.g.: Since $\frac{\mathrm{d}}{\mathrm{d}x}(-\cos x) = \sin x$, so $\int \sin(x)\,\mathrm{d}x = -\cos(x) + c$.

E.g.: Since $\frac{\mathrm{d}}{\mathrm{d}x}\frac{x^{n+1}}{n+1} = \frac{(n+1)x^n}{n+1} = x^n$, so $\int x^n\,\mathrm{d}x = \dfrac{x^{n+1}}{n+1} + c$, for $n \neq -1$.

E.g.: Since $\frac{\mathrm{d}}{\mathrm{d}x}\arcsin(x) = \frac{1}{\sqrt{1-x^2}}$, so
$$\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = \arcsin(x) + c.$$

E.g.: Since when $x > 0$, $\frac{\mathrm{d}}{\mathrm{d}x}\ln|x| = \frac{\mathrm{d}}{\mathrm{d}x}\ln x = \frac{1}{x}$ and when $x < 0$ we have $\frac{\mathrm{d}}{\mathrm{d}x}\ln|x| = \frac{\mathrm{d}}{\mathrm{d}x}\ln(-x) = \frac{-1}{-x} = \frac{1}{x}$, so $\int \dfrac{1}{x}\,\mathrm{d}x = \ln|x| + c$.

Remark: The symbol $\mathrm{d}x$ in $\int f(x)\,\mathrm{d}x$ is to indicate that x is the variable of integration. For instance $\int xy$ is ambiguous, but we know
$$\int xy\,\mathrm{d}y = \frac{1}{2}xy^2 + c \quad \text{and} \quad \int xy\,\mathrm{d}x = \frac{1}{2}x^2y + c.$$

Common integrals

- $\displaystyle\int x^n \, \mathrm{d}x = \begin{cases} \frac{x^{n+1}}{n+1} + c, & \text{if } n \neq -1 \\ \ln|x| + c, & \text{if } n = -1 \end{cases}$
- $\displaystyle\int e^x \, \mathrm{d}x = e^x + c$
- $\displaystyle\int \cos(x) \, \mathrm{d}x = \sin(x) + c$
- $\displaystyle\int \sin(x) \, \mathrm{d}x = -\cos(x) + c$
- $\displaystyle\int \sec^2(x) \, \mathrm{d}x = \tan(x) + c$
- $\displaystyle\int \frac{1}{1+x^2} \, \mathrm{d}x = \arctan(x) + c$
- $\displaystyle\int a^x \, \mathrm{d}x = \int e^{x \ln a} \, \mathrm{d}x = \frac{e^{x \ln a}}{\ln(a)} + c = \frac{a^x}{\ln(a)} + c,$ for $a > 0$ and $a \neq 1$.

Activity 1

Find the following indefinite integrals.

(a) $\displaystyle\int 5 \, \mathrm{d}x.$

(b) $\displaystyle\int t \, \mathrm{d}t.$

(c) $\displaystyle\int \sqrt{z} \, \mathrm{d}z.$

(d) $\displaystyle\int \frac{1}{t} \, \mathrm{d}t.$

(e) $\displaystyle\int \frac{1}{t^3} \, \mathrm{d}t.$

(f) $\displaystyle\int \cos x \, \mathrm{d}x.$

Activity 1: solution

(a) $\displaystyle\int 5\,\mathrm{d}x = 5x + c.$

(b) $\displaystyle\int t\,\mathrm{d}t = \frac{1}{2}t^2 + c.$

(c) $\displaystyle\int \sqrt{z}\,\mathrm{d}z = \int z^{\frac{1}{2}}\,\mathrm{d}z = \frac{1}{\frac{3}{2}}z^{\frac{3}{2}} + c = \frac{2}{3}z^{\frac{3}{2}} + c.$

(d) $\displaystyle\int \frac{1}{t}\,\mathrm{d}t = \ln|t| + c.$

(e) $\displaystyle\int \frac{1}{t^3}\,\mathrm{d}t = \int t^{-3}\,\mathrm{d}t = -\frac{1}{2}t^{-2} + c = -\frac{1}{2t^2} + c.$

(f) $\displaystyle\int \cos x\,\mathrm{d}x = \sin x + c.$

Linearity of indefinite integrals

Suppose $F'(x) = f(x)$ and $G'(x) = g(x)$, then the derivative of $\lambda\, F(x) + \mu\, G(x)$ is $\lambda\, f(x) + \mu\, g(x)$, hence

$$\int (\lambda\, f(x) + \mu\, g(x))\ \mathrm{d}x = \lambda\, F(x) + \mu\, G(x) + c,$$

in other words,

$$\int (\lambda\, f(x) + \mu\, g(x))\ \mathrm{d}x = \lambda \int f(x)\,\mathrm{d}x + \mu \int g(x)\,\mathrm{d}x.$$

This is called the linearity property of integration. Therefore, most of the time it is easier to split a long integral into a sum of short ones. It is commonly the first step when computing an integral.

Linearity of indefinite integrals - example

E.g.: Suppose we want to calculate

$$\int \frac{x^3 + 5x^2 - 4x + 1}{x^2} \mathrm{d}x.$$

We can split the fraction and integrate term by term:

$$\begin{aligned}\int \frac{x^3 + 5x^2 - 4x + 1}{x^2} \mathrm{d}x &= \int \left(x + 5 - \frac{4}{x} + \frac{1}{x^2} \right) \mathrm{d}x \\ &= \frac{x^2}{2} + 5x - 4\ln|x| - \frac{1}{x} + c.\end{aligned}$$

Linearity of indefinite integrals - example

E.g.: Suppose we want to calculate

$$\int \sin^3 x \, \mathrm{d}x.$$

Using trigonometric identities and the double angle formulas, we get

$$\begin{aligned} \sin 3\,x &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2 \sin x \cos^2 x + (1 - 2\sin^2 x) \sin x \\ &= 2 \sin x (1 - \sin^2 x) + \sin x - 2 \sin^3 x \\ &= 3 \sin x - 4\, \sin^3 x \end{aligned}$$

$$\Rightarrow \quad \sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin 3\,x.$$

So

$$\int \sin^3 x \, \mathrm{d}x = \int \left(\frac{3}{4} \sin x - \frac{1}{4} \sin 3\,x \right) \mathrm{d}x = -\frac{3}{4} \cos x + \frac{1}{12} \cos 3x + c.$$

Activity 2

Find the following indefinite integrals.

(a) $\displaystyle\int \frac{x^2+1}{x}\,\mathrm{d}x.$

(b) $\displaystyle\int x^3(4x^2-1)\,\mathrm{d}x.$

(c) $\displaystyle\int 2+3\sin x\,\mathrm{d}x.$

(d) $\displaystyle\int \frac{5}{\cos^2 x}\,\mathrm{d}x.$

(e) $\displaystyle\int \left(x^3+\frac{3}{x}\right)^2\mathrm{d}x.$

(f) $\displaystyle\int \frac{1-x^n}{1-x}\,\mathrm{d}x,$ for $n \in \mathbb{N}$..

Hint: geometric sequence.

Activity 2: solution

(a) $\displaystyle\int \frac{x^2+1}{x}\,\mathrm{d}x = \int \frac{x^2}{x} + \frac{1}{x}\,\mathrm{d}x = \int x + \frac{1}{x}\,\mathrm{d}x = \frac{1}{2}x^2 + \ln|x| + c.$

(b)
$$\int x^3(4x^2-1)\,\mathrm{d}x = \int 4x^5 - x^3\,\mathrm{d}x = \frac{4}{6}x^6 - \frac{1}{4}x^4 + c = \frac{2}{3}x^6 - \frac{1}{4}x^4 + c.$$

(c) $\displaystyle\int 2 + 3\sin x\,\mathrm{d}x = 2x - 3\cos x + c.$

(d) $\displaystyle\int \frac{5}{\cos^2 x}\,\mathrm{d}x = \int 5\sec^2 x\,\mathrm{d}x = 5\tan x + c.$

(e) Expanding $\left(x^3 + \frac{3}{x}\right)^2 = x^6 + 6x^2 + \frac{9}{x^2}$, we have

$$\int \left(x^3 + \frac{3}{x}\right)^2 \mathrm{d}x = \int \left(x^6 + 6x^2 + \frac{9}{x^2}\right) \mathrm{d}x = \frac{x^7}{7} + 2\,x^3 - \frac{9}{x} + c.$$

Activity 2: solution

(f) We use the partial sum formula of geometric sequences:

$$\frac{1-x^n}{1-x} = \sum_{k=0}^{n-1} x^k,$$

so that

$$\int \frac{1-x^n}{1-x}\,\mathrm{d}x = \int \left(\sum_{k=0}^{n-1} x^k \right) \mathrm{d}x = \sum_{k=0}^{n-1} \int x^k \,\mathrm{d}x = \sum_{k=0}^{n-1} \frac{x^{k+1}}{k+1} + c$$

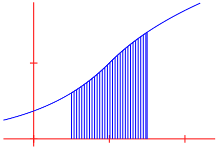
$$= \sum_{k=1}^{n} \frac{x^k}{k} + c.$$

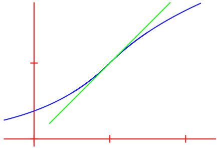
Outline

Definite integrals and signed areas

The Greeks studied the problem of finding areas under curves based on a method of exhaustion. The basic idea was to divide the area under a curve into smaller shapes each of which could be easily found.

Study of areas	Study of rates of change
$\Downarrow$	$\Downarrow$
Integral calculus	Differential calculus
(Greece 600 B.C.)	(1700 A.D.)



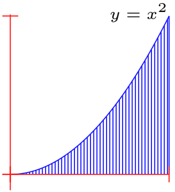


Definite integrals and signed areas

Let $f(x)$ be a function defined over an interval $[a, b]$, intuitively the definite integral $\int_a^b f(x)\,\mathrm{d}x$ is the signed area (positive if the region is above the x-axis, negative if the region is below the x-axis) of the region enclosed by the graph of $f(x)$ and the x-axis in between a and b.

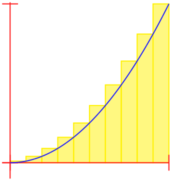
To find this area, we use the sum of areas of rectangles as an approximation and then take the limit as the number of rectangles goes to infinity. You can visualize it using this Geogebra applet.

For example suppose we want to find the area between the graph of $f(x) = x^2$ and the x-axis, over $[0, 1]$, i.e. the blue region R.

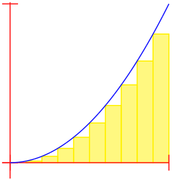


Definite integrals and signed areas

Let's subdivide the interval $[0,1]$ into n equal subintervals. For instance, if $n=10$ we have



$$\text{Area of } R \leq \frac{1}{10}\left(\frac{1}{10}\right)^2 + \frac{1}{10}\left(\frac{2}{10}\right)^2 + \cdots + \frac{1}{10}\left(\frac{10}{10}\right)^2 = 0.385.$$



$$\text{Area of } R \geq \frac{1}{10}\left(\frac{0}{10}\right)^2 + \frac{1}{10}\left(\frac{1}{10}\right)^2 + \cdots + \frac{1}{10}\left(\frac{9}{10}\right)^2 = 0.285.$$

Definite integrals and signed areas

For a general number of subintervals n, we always have

$$L_n \leq \text{Area of } R \leq U_n$$

where

$$U_n = \frac{1}{n}\left(\left(\frac{1}{n}\right)^2 + \left(\frac{2}{n}\right)^2 + \cdots + \left(\frac{n}{n}\right)^2\right),$$

$$L_n = \frac{1}{n}\left(\left(\frac{0}{n}\right)^2 + \left(\frac{1}{n}\right)^2 + \cdots + \left(\frac{n-1}{n}\right)^2\right).$$

Since the inequalities are true for all $n \in \mathbb{N}$, even when $n \to \infty$, so by taking the limit as $n \to \infty$ we get

$$\lim_{n\to\infty} L_n \leq \text{Area of } R \leq \lim_{n\to\infty} U_n.$$

If these two limits coincide, then we'll get the Area of R as the common limit value.

Let's find these limits.

$\lim_{n\to\infty} U_n$

The following theorem is known (see the appendix).

Theorem

$$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\begin{aligned}
\lim_{n\to\infty} U_n &= \lim_{n\to\infty} \frac{1}{n^3}(1^2 + 2^2 + \cdots + n^2) \\
&= \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3} \\
&= \lim_{n\to\infty} \frac{1\cdot(1+\frac{1}{n})(2+\frac{1}{n})}{6} = \frac{1}{3}.
\end{aligned}$$

$\lim_{n\to\infty} L_n$

Theorem

$$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

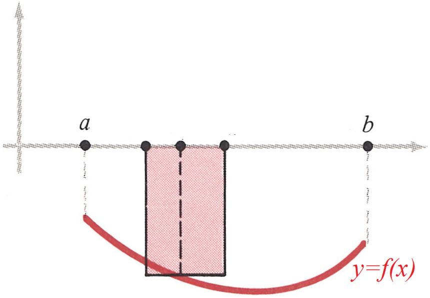
$$\begin{aligned}
\lim_{n\to\infty} L_n &= \lim_{n\to\infty} \frac{1}{n^3}(0^2 + 1^2 + 2^2 + \cdots + (n-1)^2) \\
&= \lim_{n\to\infty} \frac{(n-1)n(2n-1)}{6n^3} \\
&= \lim_{n\to\infty} \frac{(1-\frac{1}{n})\cdot 1 \cdot (2-\frac{1}{n})}{6} = \frac{1}{3}.
\end{aligned}$$

We see that $\lim_{n\to\infty} U_n = \lim_{n\to\infty} L_n = \frac{1}{3}$. Since the Area of R is in between $\lim_{n\to\infty} U_n$ and $\lim_{n\to\infty} L_n$, so the Area of R must be $\frac{1}{3}$ too.

Definite integrals and signed areas

The previous example of computing the area of the region between $f(x) = x^2$ and the x-axis over an interval, through approximations of the region by rectangles of shrinking base length, is how we generally define the area of an irregular region.

Note that if the region is below the x-axis, i.e. $f(x) < 0$, then the 'heights' of the rectangles $= f(x)$ are negative, thus resulting in a negative signed area.

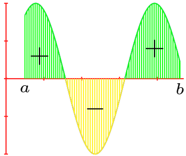


Definite integrals and areas

As the actual definition of definite integrals is quite technical, we are content with the following intuitive definition for this course. You'll learn the actual definition in Calculus in term 1.

Definition:

The ***definite integral*** of $f(x)$ with respect to x over $[a,b]$, denoted as $\int_a^b f(x)\,\mathrm{d}x$, is our intuitive idea of the 'area' of the region between the curve $y = f(x)$ and the x-axis (where 'area' for a region below the x-axis counts negative) from $x = a$ to $x = b$.



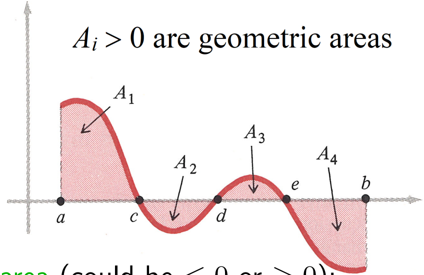
Geometric and algebraic/signed areas

Remark: As illustrated in the example of $f(x) = x^2$, finding the signed area/definite integral involves limits, which may or may not exist for a general function $f(x)$. So bear in mind that not all functions have definite integrals. For those functions whose definite integrals are defined/exist, we called them ***integrable***.

Definition:

If $f(x)$ is integrable, we define the value $\int_a^b |f(x)|\,\mathrm{d}x$ to be the ***(geometric) area*** between the curve $y = f(x)$ and the x-axis over $[a, b]$. On the other hand, $\int_a^b f(x)\,\mathrm{d}x$ is called the ***signed area*** or ***algebraic area*** between the curve $y = f(x)$ and the x-axis over $[a, b]$.

Geometric and algebraic/signed areas



Algebraic/signed area (could be ≤ 0 or ≥ 0):

$$\int_a^b f(x)\mathrm{d}x = A_1 - A_2 + A_3 - A_4, \text{ where } A_1, A_2, A_3, A_4 \text{ are positive.}$$

Geometric area or simply area (always ≥ 0):

$$A_1 + A_2 + A_3 + A_4 = \int_a^c f(x)\mathrm{d}x - \int_c^d f(x)\mathrm{d}x + \int_d^e f(x)\mathrm{d}x - \int_e^b f(x)\mathrm{d}x$$

$$= \int_a^b |f(x)|\mathrm{d}x.$$

Examples of integrable functions

Theorem

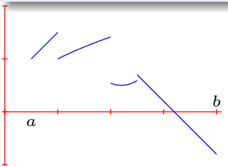
If $f(x)$ is a continuous function on $[a,b]$, then $f(x)$ is integrable.

Definition:

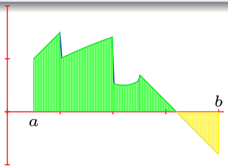
A function is ***piecewise continuous*** on a closed interval $[a,b]$ if the function is continuous everywhere on $[a,b]$ except for a finite number of points where we have jump discontinuities.

Theorem

Piecewise continuous functions on $[a,b]$ are integrable.



$\Rightarrow$



Continuity on a closed interval implies integrability, but integrability does not imply continuity!

Computing definite integrals

Definition:

If we have $\int f(x)\,\mathrm{d}x$ or $\int_a^b f(x)\,\mathrm{d}x$, the function $f(x)$ is called the ***integrand***. For the latter, a is called the ***lower limit*** and b is called the ***upper limit*** of the definite integral.

How do we find the value of $\int_a^b f(x)\,\mathrm{d}x$?

At this stage we need to use areas of rectangles with shrinking base lengths, which involves limits.

Next class we will learn a powerful theorem that allows us to compute definite integrals using antiderivatives: the Fundamental Theorem of Calculus.

Activity 3

Find the following definite integral by mimicking the steps of the example on slide 19.

$$\int_0^b x^3 \, \mathrm{d}x$$

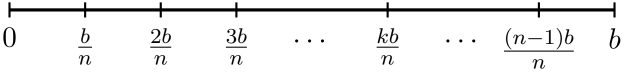
where $b > 0$.

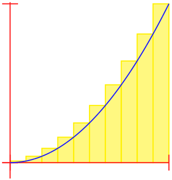
Hint: you could use the following formula.

$$\begin{aligned} 1^3 + 2^3 + \ldots + n^3 &= (1 + 2 + \ldots + n)^2 \\ &= \frac{n^2(n+1)^2}{4}. \end{aligned}$$

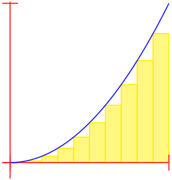
Activity 3: solution

Let the region between $y = x^3$ and the x-axis over the interval $[0, b]$ be R. We divide the interval $[0, b]$ into n equal subintervals of length $\Delta x = \frac{b}{n}$:





$$\text{Signed area of } R \leq \frac{b}{n}\left(\frac{b}{n}\right)^3 + \frac{b}{n}\left(\frac{2b}{n}\right)^3 + \cdots + \frac{b}{n}\left(\frac{nb}{n}\right)^3 =: U_n.$$



$$\text{Signed area of } R \geq \frac{b}{n}\left(\frac{0 \cdot b}{n}\right)^3 + \frac{b}{n}\left(\frac{b}{n}\right)^3 + \cdots + \frac{b}{n}\left(\frac{(n-1)b}{n}\right)^3 =: L_n.$$

Activity 3: solution

For all $n \in \mathbb{N}$, we always have

$$L_n \leq \text{Signed area of } R = \int_0^b x^3 \, \mathrm{d}x \leq U_n.$$

So when $n \to \infty$, we have

$$\lim_{n\to\infty} L_n \leq \int_0^b x^3 \, \mathrm{d}x \leq \lim_{n\to\infty} U_n.$$

Moreover,

$$\begin{aligned}
\lim_{n\to\infty} U_n &= \lim_{n\to\infty} \left(\frac{b}{n}\left(\frac{b}{n}\right)^3 + \frac{b}{n}\left(\frac{2b}{n}\right)^3 + \cdots + \frac{b}{n}\left(\frac{nb}{n}\right)^3 \right) \\
&= \lim_{n\to\infty} \frac{b}{n} \left(\frac{1^3 b^3}{n^3} + \frac{2^3 b^3}{n^3} + \cdots + \frac{n^3 b^3}{n^3} \right) \\
&= \lim_{n\to\infty} \frac{b}{n} \cdot \frac{b^3}{n^3} \cdot \left(1^3 + 2^3 + \cdots + n^3\right) \\
&= \lim_{n\to\infty} \frac{b^4}{n^4} \frac{n^2(n+1)^2}{4} = \frac{b^4}{4} \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^2 = \frac{b^4}{4}(1+0)^2 = \frac{b^4}{4}.
\end{aligned}$$

Activity 3: solution

$$
\begin{aligned}
\lim_{n\to\infty} L_n &= \lim_{n\to\infty}\left(\frac{b}{n}\left(\frac{0\cdot b}{n}\right)^3+\frac{b}{n}\left(\frac{2b}{n}\right)^3+\cdots+\frac{b}{n}\left(\frac{(n-1)b}{n}\right)^3\right)\\
&= \lim_{n\to\infty}\frac{b}{n}\left(\frac{0^3b^3}{n^3}+\frac{1^3b^3}{n^3}+\cdots+\frac{(n-1)^3b^3}{n^3}\right)\\
&= \lim_{n\to\infty}\frac{b}{n}\cdot\frac{b^3}{n^3}\cdot\left(0^3+1^3+\cdots+(n-1)^3\right)\\
&= \lim_{n\to\infty}\frac{b^4}{n^4}\frac{(n-1)^2((n-1)+1)^2}{4}\\
&= \frac{b^4}{4}\lim_{n\to\infty}\left(\frac{n-1}{n}\right)^2=\frac{b^4}{4}\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^2=\frac{b^4}{4}(1-0)^2=\frac{b^4}{4}.
\end{aligned}
$$

Since

$$
\frac{b^4}{4}=\lim_{n\to\infty}L_n\le\int_0^b x^3\,\mathrm{d}x\le\lim_{n\to\infty}U_n=\frac{b^4}{4},
$$

we must have $\displaystyle\int_0^b x^3\,\mathrm{d}x=\frac{b^4}{4}$.

Outline

Properties of definite integrals

We will list down some important and useful properties of definite integrals in this section. All the properties have a geometrical interpretation in terms of signed areas. Understanding these geometrical interpretations will help you to remember the properties without any effort.

Properties of definite integrals

Proposition

For any integrable functions $f(x), g(x)$ and any constants λ, μ, we have

1. Constant multiple rule:

$$\int_a^b \lambda f(x)\,\mathrm{d}x = \lambda \int_a^b f(x)\,\mathrm{d}x.$$

2. Addition rule:

$$\int_a^b (f(x) + g(x))\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + \int_a^b g(x)\,\mathrm{d}x$$

3. Linearity:

$$\int_a^b (\lambda f(x) + \mu g(x))\,\mathrm{d}x = \lambda \int_a^b f(x)\,\mathrm{d}x + \mu \int_a^b g(x)\,\mathrm{d}x$$

Properties of definite integrals

Proposition

4. Addition with respect to intervals:

$$\int_a^{b} f(x)\,\mathrm{d}x + \int_{b}^c f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x.$$

5. $$\int_a^a f(x)\,\mathrm{d}x = 0.$$

So far, we only define the definite integral $\int_a^b f(x)\,\mathrm{d}x$ for $a \leq b$.

Definition

If $a > b$, then we define

$$\int_a^b f(x)\,\mathrm{d}x := -\int_b^a f(x)\,\mathrm{d}x.$$

Activity 4

(a) Compute $\int_{6.5}^{9} f(x)\,\mathrm{d}x$ if

$$\int_{4}^{11.5} 2f(x)\,\mathrm{d}x = 8, \quad \int_{6.5}^{4} f(x)\,\mathrm{d}x = -3, \quad \int_{9}^{11.5} \frac{1}{2} f(x)\,\mathrm{d}x = 2.$$

(b) Compute

$$\int_{2}^{2} \frac{1}{t+t^2}\,\mathrm{d}t.$$

(c) By using the results from slide 23 and activity 3, compute

$$\int_{0}^{1} (-x^3 + 4x^2)\,\mathrm{d}x.$$

Activity 4: solution

(a) The given information implies

$$\int_4^{11.5} f(x)\,\mathrm{d}x = \frac{1}{2}\int_4^{11.5} 2f(x)\,\mathrm{d}x = \frac{8}{2} = 4,$$
$$\int_4^{6.5} f(x)\,\mathrm{d}x = -\int_{6.5}^{4} f(x)\,\mathrm{d}x = -(-3) = 3,$$
$$\int_9^{11.5} f(x)\,\mathrm{d}x = 2\int_9^{11.5} \frac{1}{2}f(x)\,\mathrm{d}x = 2\times 2 = 4.$$

Since

$$\int_4^{6.5} f(x)\,\mathrm{d}x + \int_{6.5}^{9} f(x)\,\mathrm{d}x + \int_9^{11.5} f(x)\,\mathrm{d}x = \int_4^{11.5} f(x)\,\mathrm{d}x,$$

so

$$\int_{6.5}^{9} f(x)\,\mathrm{d}x = \int_4^{11.5} f(x)\,\mathrm{d}x - \int_4^{6.5} f(x)\,\mathrm{d}x - \int_9^{11.5} f(x)\,\mathrm{d}x = -3.$$

Activity 4: solution

(b)

$$\int_2^2 \frac{1}{t+t^2}\,\mathrm{d}t = 0.$$

(c) We have $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$ from slide 23 and $\int_0^b x^3\,\mathrm{d}x = \frac{b^4}{4}$ from activity 3, hence by linearity, we have

$$\begin{aligned}\int_0^1 (-x^3+4x^2)\,\mathrm{d}x &= -\int_0^1 x^3\,\mathrm{d}x + 4\int_0^1 x^2\,\mathrm{d}x \\ &= -\frac{1^4}{4} + 4\cdot\frac{1}{3} \\ &= \frac{13}{12}.\end{aligned}$$

Properties of definite integrals

Proposition

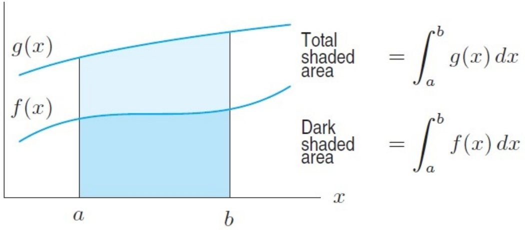
6. If $f(x) > 0$ for $a < x < b$, then

$$\int_a^b f(x)\,\mathrm{d}x > 0.$$

7. Monotonicity: If $f(x) \le g(x)$ for $a < x < b$, then

$$\int_a^b f(x)\,\mathrm{d}x \le \int_a^b g(x)\,\mathrm{d}x.$$

You'll find them trivial if you interpret them geometrically!



Properties of definite integrals - example

E.g.: Since $f(x) = e^{-x^2} > 0$ over $\mathbb{R}$, so

$$\int_{-3}^{5} e^{-x^2}\,\mathrm{d}x > 0.$$

E.g.: Since $\sin(x^2) \leq 1$ for all $x \in \mathbb{R}$, so we have

$$\int_0^{\sqrt{\pi}} \sin(x^2)\,\mathrm{d}x \leq \int_0^{\sqrt{\pi}} 1\,\mathrm{d}x.$$

Furthermore, $\displaystyle\int_0^{\sqrt{\pi}} 1\,\mathrm{d}x$ is geometrically the signed area between the graph $y = 1$ and the x-axis, over the interval $[0, \sqrt{\pi}]$, i.e. the area of a rectangle of height 1 and length $\sqrt{\pi}$, so $\displaystyle\int_0^{\sqrt{\pi}} 1\,\mathrm{d}x = 1 \times \sqrt{\pi} = \sqrt{\pi}$. Hence,

$$\int_0^{\sqrt{\pi}} \sin(x^2)\,\mathrm{d}x \leq \sqrt{\pi}.$$

Properties of definite integrals

Proposition

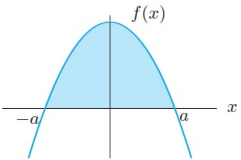
8. If $g(x)$ is an odd function, then

$$\int_{-a}^{a} g(x)\,\mathrm{d}x = 0.$$

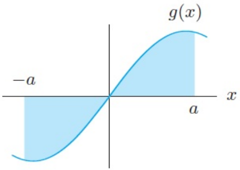
9. If $f(x)$ is an even function, then for $a \geq 0$

$$\int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_{0}^{a} f(x)\,\mathrm{d}x.$$

Again, you'll find them trivial if you interpret them geometrically!



For an even function, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



For an odd function, $\int_{-a}^{a} g(x)\,dx = 0$

Properties of definite integrals - example

E.g.: Since $f(x) = 6x^5$ is an odd function as $f(-x) = 6(-x)^5 = -6x^5 = -f(x)$, so for any constant $a \in \mathbb{R}$,

$$\int_{-a}^{a} 6x^5 \,\mathrm{d}x = 0.$$

E.g.: Since $g(x) = x^2$ is an even function because $g(-x) = (-x)^2 = x^2$ and $h(x) = x\sqrt{x^6 + x^4 + x^2 + 1}$ is an odd function because

$$\begin{aligned} h(-x) &= (-x)\sqrt{(-x)^6 + (-x)^4 + (-x)^2 + 1} \\ &= -x\sqrt{x^6 + x^4 + x^2 + 1} = -h(x), \end{aligned}$$

so

$$\begin{aligned} &\int_{-1}^{1} \left(x^2 - x\sqrt{x^6 + x^4 + x^2 + 1}\right) \mathrm{d}x \\ &= \int_{-1}^{1} x^2 \,\mathrm{d}x - \int_{-1}^{1} \left(x\sqrt{x^6 + x^4 + x^2 + 1}\right) \mathrm{d}x \\ &= 2\int_{0}^{1} x^2 \,\mathrm{d}x - 0 = 2 \cdot \frac{1}{3} = \frac{2}{3}. \end{aligned}$$

Outline

Summary

- Antiderivatives and Indefinite Integrals
  - definition of antiderivatives and indefinite integrals.
  - common integrals.
  - linearity of indefinite integrals.
- Signed Areas and Definite Integrals
  - definite integrals as signed areas, integrable functions.
  - algebraic/signed areas versus geometric areas.
  - computing definite integrals by limits.
  - continuous and piecewise continuous functions are integrable.
  - integrand, lower and upper limits.
- Properties of Definite Integrals
  - constant multiple rule, addition rule, linearity, addition with respect to intervals.
  - definition of definite integrals $\int_a^b f(x)\,\mathrm{d}x$ for $a > b$.
  - monotonicity properties of definite integrals, definite integrals of odd/even functions.

Resources

References:

- George F. Simmons. Calculus with Analytic Geometry (2nd edition) sections 5.3, 6.1-6.5.
- Hughes-Hallett. Calculus (6th edition) sections 6.1, 6.2, 5.1, 5.2, 5.4.

Outline

$1^2 + 2^2 + 3^2 + \cdots + n^2$

Theorem

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

Proof: $(k+1)^3 - k^3 = (k^3 + 3k^2 + 3k + 1) - k^3 = 3k^2 + 3k + 1.$

Therefore,

$$\begin{array}{rcl}
2^3 - 1^3 & = & 3(1^2) + 3(1) + 1 \\
3^3 - 2^3 & = & 3(2^2) + 3(2) + 1 \\
4^3 - 3^3 & = & 3(3^2) + 3(3) + 1 \\
\cdots & = & \cdots \\
+ \quad (n+1)^3 - n^3 & = & 3(n^2) + 3(n) + 1 \\
\hline
(n+1)^3 - 1^3 & = & 3(1^2 + 2^2 + \cdots + n^2) + 3(1 + 2 + \cdots + n) + n
\end{array}$$

$$\Rightarrow \quad \sum_{k=1}^{n} k^2 = \frac{(n+1)^3 - 1^3 - n - 3\sum_{k=1}^{n} k}{3} = \frac{n^3 + 3n^2 + 3n + 1 - 1 - n - 3\frac{n(n+1)}{2}}{3}$$

$$= \frac{2n^3 + 3n^2 + n}{6} = \frac{n(n+1)(2n+1)}{6}$$

Q.E.D.

$1^3 + 2^3 + 3^3 + \cdots + n^3$

Theorem

$$\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$$

Proof:

$(k+1)^4 - k^4 = (k^4 + 4k^3 + 6k^2 + 4k + 1) - k^3 = 4k^3 + 6k^2 + 4k + 1.$

Therefore,

$$\begin{array}{rcl}
2^4 - 1^4 & = & 4(1^3) + 6(1^2) + 4 \cdot 1 + 1 \\
3^4 - 2^4 & = & 4(2^3) + 6(2^2) + 4 \cdot 2 + 1 \\
4^4 - 3^4 & = & 4(3^3) + 6(3^2) + 4 \cdot 3 + 1 \\
\cdots & = & \cdots \\
+ \quad (n+1)^4 - n^4 & = & 4(n^3) + 6(n^2) + 4 \cdot n + 1 \\
\hline
(n+1)^4 - 1^4 & = & 4\left(\sum_{k=1}^{n} k^3\right) + 6\left(\sum_{k=1}^{n} k^2\right) + 4\left(\sum_{k=1}^{n} k\right) + n
\end{array}$$

$$\Rightarrow \quad \sum_{k=1}^{n} k^3 = \frac{(n+1)^4 - 1^4 - n - 4\sum_{k=1}^{n} k - 6\sum_{k=1}^{n} k^2}{4}$$

$$= \frac{n^4 + 4n^3 + 6n^2 + 4n - n - 4\frac{n(n+1)}{2} - 6\frac{n(n+1)(2n+1)}{6}}{4}$$

$$= \frac{n^4 + 2n^3 + n^2}{4} = \frac{n^2(n+1)^2}{4}$$

Q.E.D.